

Effects of chlorophyll and chlorophyllin on low-dose aflatoxin B(1) pharmacokinetics in human volunteers.

Chlorophyll (Chla) and chlorophyllin (CHL) were shown previously to reduce carcinogen bioavailability, biomarker damage, and tumorigenicity in trout and rats. These findings were partially extended to humans, where CHL reduced excretion of aflatoxin B(1) (AFB(1))-DNA repair products in Chinese unavoidably exposed to dietary AFB(1). However, neither AFB(1) pharmacokinetics nor Chla effects were examined. We conducted an unblinded crossover study to establish AFB(1) pharmacokinetic parameters among four human volunteers, and to explore possible effects of CHL or Chla cotreatment in three of those volunteers. For protocol 1, fasted subjects received an Institutional Review Board-approved dose of ^{14}C -AFB(1) (30 ng, 5 nCi) by capsule with 100 mL water, followed by normal eating and drinking after 2 hours. Blood and cumulative urine samples were collected over 72 hours, and ^{14}C -AFB(1) equivalents were determined by accelerator mass spectrometry. Protocols 2 and 3 were similar except capsules also contained 150 mg of purified Chla or CHL, respectively. Protocols were repeated thrice for each volunteer. The study revealed rapid human AFB(1) uptake (plasma $k(a)$, $5.05 \pm 1.10 \text{ h}^{-1}$; $T(\text{max})$, 1.0 hour) and urinary elimination (95% complete by 24 hours) kinetics. Chla and CHL treatment each significantly impeded AFB(1) absorption and reduced C_{max} and AUCs (plasma and urine) in one or more subjects. These initial results provide AFB(1) pharmacokinetic parameters previously unavailable for humans, and suggest that Chla or CHL co-consumption may limit the bioavailability of ingested aflatoxin in humans, as they do in animal models.